PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – V

COMPOSITE BEAMS

Syllabus : Composite beams – different types – propped and un-propped – stress distribution – differential shrinkage – analysis of composite beams.

Contents

1. Introduction & advantages of composite construction
2. Types of composite beams
3. Propped & un-propped construction
4. Stages of loading and stress analysis
5. Properties of composite section (effective width)
6. Worked example – un-propped construction
7. Worked example – propped construction
8. Differential shrinkage – theory
9. Differential shrinkage – example
10. Horizontal (interface) shear & shear connectors
11. Ultimate strength of composite beams
12. Important questions & formula sheet

Exam tip : The standard question is "analyse the composite beam for the three stages and find the stresses" – draw the stress diagrams for each stage and add them.

1. Introduction to Composite Construction

A **composite beam** consists of a **precast prestressed beam** and a **cast-in-situ concrete slab (or topping)** acting together as one unit through bond / shear connectors (stirrups projecting from the beam).

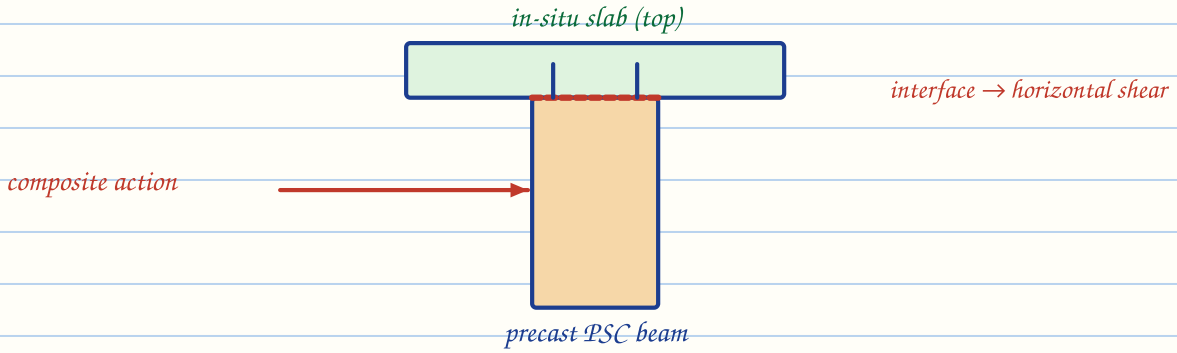


Fig. Composite section – precast beam + in-situ slab (shear ties across interface)

Why composite ?

- Precast beams are made in factory (pre-/post-tensioned) – good quality, quick erection; slab cast in-situ over them gives the deck (bridges, buildings, floors).
- The slab acts as a **compression flange** – increases the section modulus and moment of resistance → lesser prestress force and steel.
- Precast beam acts as the **formwork / support** for the slab (saving in shuttering – especially in bridges).
- Continuity of the deck can be provided by the in-situ concrete; monolithic appearance.
- Speed of construction, economy – the part of load (self-weight) is carried by the precast section alone.

Limitations

- Differential shrinkage and creep between precast and in-situ concrete causes additional stresses.
- Interface shear has to be provided for carefully (roughening, stirrups).
- The slab concrete is of lower grade (M25–M30) than the precast beam (M40–M60).
- Composite action starts only after the in-situ concrete has hardened – **un-propped** beams carry slab weight alone.

2. Types of Composite Beams

Type	Description
1. Precast PSC beam + cast-in-situ RC slab	Most common (T-beam bridge / floor). Beam pre-/post-tensioned; slab 100–200 mm.
2. Precast PSC beam + precast slab (with in-situ joints)	Fully precast deck, composite after grouting/screeding.
3. Precast RC/PSC beam + in-situ topping on hollow-core / ribbed slabs	Floors in buildings; topping 40–75 mm.
4. In-situ PSC + in-situ slab (two-stage casting)	Beam prestressed first; slab cast afterwards; long span.
5. Steel beam + concrete slab (steel–concrete composite)	Not prestressed generally; also possible with prestressing.
6. Post-tensioned precast + continuity (stitching)	Precast girders made continuous with in-situ crosshead / deck prestressing.

Common cross-sections

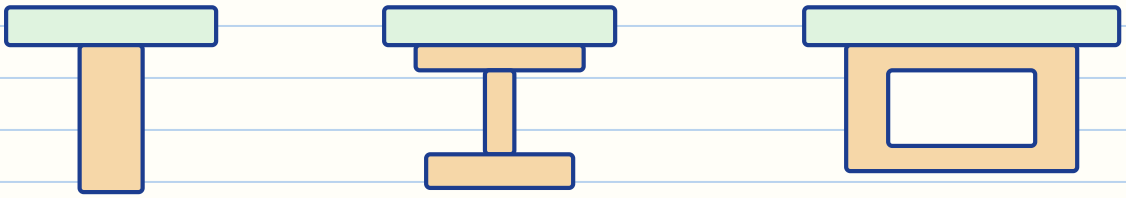


Fig. Typical composite sections (green = in-situ slab)

Based on construction method

- **Un-propped** – the precast beam carries its own weight + wet in-situ slab + formwork alone; composite action for later loads only.
- **Propped** – the beam is supported by props (temporary supports) while the in-situ slab is cast; the props are removed after the slab gains strength → the **composite section carries the whole weight**.

3. Propped vs Un-propped Construction

Aspect	Un-propped	Propped
Slab weight carried by	Precast beam alone (section 1)	Composite section (after props removed)
Props	None – simple, cheaper, faster	Props / shores required – costly, slow
Precast beam stress under slab weight	High (beam bends under wet slab)	Low – beam only carries own weight
Section needed	Stronger / deeper precast beam	Lighter precast beam possible
Total prestress required	Larger	Smaller
Deflection	Larger (before composite action)	Smaller
Suitability	Bridges over rivers/roads, high clearance	Buildings, low heights, short spans
Final stress in beam bottom	Lower compressive (more tension)	More favourable

(a) UN-PROPPED : slab load W_s on precast beam only



(b) PROPPED : props carry slab load until it hardens



Fig. Un-propped and propped construction

Un-propped : Stage 2 loads (slab weight) act on the **precast section**. Propped : slab weight acts on the **composite section** when the props are removed.

4. Stages of Loading and Stress Distribution

Stress at any fibre = sum of the stresses of all stages, each computed with the **section that actually resists it**.

Stage	Loads	Section resisting	Prestress
1. Transfer	$P_i + M_{g1}$ (beam self-weight)	Precast beam (A_1, I_1)	P_i
2. Casting of slab (un-propped)	$+ M_s$ (wet slab, formwork)	Precast beam alone	P_e (after losses)
3. After slab hardened	$+ M_q$ (finishes, live load, super-imposed)	**Composite** section (A_c, I_c)	P_e

Stress equations (compression +ve)

$$\text{Stage 1/2 (precast, unpropped)}: f = P_e/A_1 \mp P_e e/Z_1 \pm (M_{g1} + M_s)/Z_1$$

$$\text{Stage 3 (composite)}: f = \pm M_q / Z_c \text{ (at the level considered)}$$

Final stress = Stage 2 stress + Stage 3 stress (superposition)

- **At the interface**, the precast beam top fibre and slab bottom fibre have different stage stresses; but the strain is continuous after composite action (for the same E_c).
- The slab carries only the **stage-3** stress (and shrinkage effects) – none of the prestress.
- If E_c of slab ≠ E_c of beam, use modular ratio $m = E_{slab}/E_{beam}$ to transform the slab width : $b_{transformed} = m \cdot b_f$.

Effective flange width (IS 456 cl. 23.1.2)

$$b_f = L_0/6 + b_w + 6\,D_f \text{ (T-beam)}\ ;\ \le \text{spacing of beams}$$

L_0 = distance between points of zero moment (= span for simply supported beam), b_w = web width, D_f = slab thickness.

5. Worked Example – Un-propped Beam (data)

A precast prestressed rectangular beam 250 mm × 500 mm, simply supported, span 10 m, supports an in-situ slab 1000 mm × 100 mm (beam spacing 1 m; the code effective width $L/6 + b_w + 6D_f$ = 2517 mm > 1000, so 1000 mm governs). Construction un-propped. Same grade of concrete (m = 1).

Data	Value
Prestress after transfer P_i	700 kN at e = 130 mm below C.G. of the precast beam
Effective prestress $P_e = 0.8\ P_i$	560 kN
Self-weight of beam	0.25 × 0.5 × 25 = 3.125 kN/m
Weight of wet slab	1.0 × 0.1 × 25 = 2.5 kN/m
Imposed (finishes + live) on composite	7 kN/m

Moments (L = 10 m, $wL^2/8 = 12.5\ w$)

$M_{g1} = 3.125 \times 12.5 = 39.06$ kN·m ; $M_s = 2.5 \times 12.5 = 31.25$ kN·m ; $M_q = 7 \times 12.5 = 87.5$ kN·m

Section properties

Precast beam alone (section 1) :

$A_1 = 125000\ mm^2$; $I_1 = 250 \times 500^3/12 = 2.604 \times 10^9\ mm^4$; $Z_1 = I_1/250 = 1.0417 \times 10^7\ mm^3$

Composite section (section 2) – slab 1000 × 100 on top of beam :

$A_c = 125000 + 100000 = 225000\ mm^2$

$\bar{y}_b = (125000 \times 250 + 100000 \times 550)/225000 = 383.3$ mm above the beam bottom ; $\bar{y}_t = 600 - 383.3 = 216.7$ mm below slab top

$I_c = 2.604 \times 10^9 + 125000 \times 133.3^2 + 8.333 \times 10^7 + 100000 \times 166.7^2 = 7.687 \times 10^9\ mm^4$

y at beam bottom = 383.3 ; at interface (beam top) = 116.7 above C.G. ; at slab top = 216.7 above C.G.

6. Stress Analysis – Un-propped

Stage 1 – at transfer (precast alone, $P_i + M_{g1}$)

$$P_i/A_1 = 5.60 \; ; \; P_i e/Z_1 = 700\times10^3\times130/1.0417\times10^7 = 8.74 \; ; \; M_{g1}/Z_1 = 3.75 \; (N/mm^2)$$

$$\text{Top} = 5.60 - 8.74 + 3.75 = +0.61 \; ; \; \text{Bottom} = 5.60 + 8.74 - 3.75 = 10.59 \; ✓$$

Stage 2 – slab cast (precast alone, $P_e + M_{g1} + M_s$)

$$P_e/A_1 = 4.48 \; ; \; P_e e/Z_1 = 6.99 \; ; \; (M_{g1} + M_s)/Z_1 = 70.31\times10^6/1.0417\times10^7 = 6.75$$

$$\text{Top (beam)} = 4.48 - 6.99 + 6.75 = 4.24 \; ; \; \text{Bottom} = 4.48 + 6.99 - 6.75 = 4.72$$

Stage 3 – imposed load on composite section ($M_q = 87.5$ kN·m)

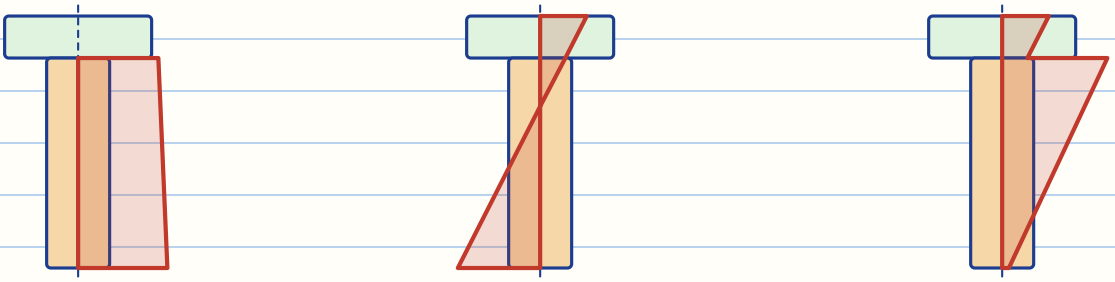
Fibre	y from C.G. (mm)	$M_q y/I_c$ (N/mm²)
Slab top	216.7 ↑	$87.5\times10^6\times216.7/7.687\times10^9$ = **+2.47 C**
Slab bottom / beam top (interface)	116.7 ↑	**+1.33 C**
Beam bottom	383.3 ↓	**−4.36 T**

Final stresses (N/mm²)

Fibre	Stage 2	Stage 3	Final
Slab top	–	+2.47	**2.47 C**
Slab bottom (interface)	–	+1.33	**1.33 C**
Beam top (interface)	+4.24	+1.33	**5.57 C**
Beam bottom	+4.72	−4.36	**0.36 C**

Beam bottom is still in compression (0.36 N/mm²) – **no tension** ✓. Permissible compression at beam top 5.57 ≪ 16 N/mm² ✓.

Stress diagrams (un-propped)



Stage 2 : $P_e + M_{g1} + M_s$ (beam only) | Stage 3 : M_q (composite) (composite section) | Final (sum of stages)

Fig. Stress distribution – stage-wise and final (red area = stress, right = compression)

- In stage 3 the stress diagram is **one straight line** over the whole composite depth (plane sections remain plane).
- Final diagram has a **jump** at the interface – the slab does not carry prestress / the stage-2 stresses.

7. Worked Example – Propped Construction (same data)

Props carry the slab weight → precast beam alone carries $P_e + M_{g1}$. After removing the props, the composite section carries $M_s + M_q = 31.25 + 87.5 = 118.75$ kN·m.

Stage 2 (beam alone, $P_e + M_{g1}$) : Top = 4.48 – 6.99 + 3.75 = 1.24 ; Bottom = 4.48 + 6.99 – 3.75 = 7.72

Stage 3 (composite, $M = 118.75$) : $M/I_c = 118.75\times10^6 / 7.687\times10^8 = 0.01545$ N/mm³ ; stress = $0.01545 \times y$ →

Fibre	Stage 2	Stage 3	Final
Slab top	–	0.01545 × 216.7 = +3.35	**3.35 C**
Slab bottom	–	+1.80 (y = 116.7)	**1.80 C**
Beam top	+1.24	+1.80	**3.04 C**
Beam bottom	+7.72	−5.92 (y = 383.3)	**1.80 C**

Propped : beam bottom = 1.80 N/mm² (more compression) vs 0.36 N/mm² un-propped → propping improves the stress state and allows lesser P.

8. Differential Shrinkage

The precast beam is old (its shrinkage is mostly complete), whereas the in-situ slab is new and shrinks a lot after casting. The beam **restrains** the slab → stresses develop in the composite beam.

- The slab tends to shrink more → tensile stress in the slab (restrained), compression + tension in beam; a **sagging curvature** (deflection) is produced.
- **Differential shrinkage strain** $\varepsilon_{ds} = \varepsilon_{cs}$ (slab) $- \varepsilon_{cs}$ (precast) $\approx 1.5\times10^{-4}$ to 3×10^{-4} (IS 1343 : shrinkage strain 0.0003 pre – 0.0002/log(t+2) post).
- Creep of concrete **relieves** part of the stress – use the effective modulus $E_{ce} = E_c/(1+\theta)$ for the slab.

Analysis procedure (3 steps)

1. **Free shrinkage** : the slab is allowed to shrink freely – no stress.
2. **Restrain** the slab : apply a tensile force at the slab C.G. to bring it back to the original length : $N = \varepsilon_{ds} \cdot E_{ce} \cdot A_s$ (A_s = slab area) → tensile stress in slab = $N/A_s = \varepsilon_{ds} E_{ce}$.
3. **Release** the restraint : apply an equal and opposite force N (compressive) at the slab C.G. to the **composite section** – it acts at eccentricity e_s from the composite centroid.

$$\text{Stress from release : } f = N/A_c \pm N e_s y / I_c$$

$$\text{Final slab stress} = -N/A_s \text{ (restraint tension)} + N/A_c + N e_s y/I_c$$

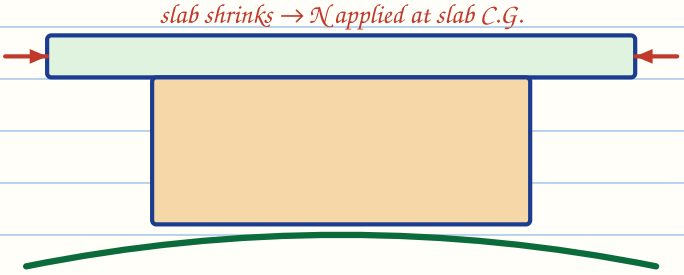


Fig. Differential shrinkage – restraint force and curvature

9. Differential Shrinkage – Example

For the composite beam of Example 5 : slab 1000 × 100, beam 250 × 500, E_c = 36000 N/mm², differential shrinkage strain $\varepsilon_{ds} = 2.0\times10^{-4}$, creep coefficient θ = 1.6. Find the stresses due to differential shrinkage.

$$E_{ce} = 36000/(1 + 1.6) = 13846\ N/mm^2$$

$$N = \varepsilon_{ds} E_{ce} A_s = 2\times10^{-4} \times 13846 \times 100000 = 276.9\ kN \text{ (slab tension if restrained} = 2.77\ N/mm^2)$$

$$e_s = 550 - 383.3 = 166.7\ mm\ ;\ N e_s = 276.9\times10^3 \times 166.7 = 4.615\times10^7\ N{\cdot}mm$$

$$N/A_c = 276900/225000 = 1.23\ N/mm^2$$

Fibre	N/A_c	$N e_s y/I_c$	Release stress	With slab restraint (−2.77)
Slab top	+1.23	+1.30 (y = 216.7 ↑)	+2.53 C	−0.24 (T)
Slab bottom	+1.23	+0.70	+1.93 C	−0.84 (T)
Beam top	+1.23	+0.70	+1.93 C	–
Beam bottom	+1.23	−2.30 (y = 383.3 ↓)	**−1.07 (T)**	–

- *The slab remains nearly stress free; the beam bottom receives a **tension of 1.07 N/mm²** which is subtracted from the earlier 0.36 C : final = 0.36 − 1.07 = **−0.71 N/mm² (tension)** → small tension; acceptable for Type 2 (limited tension) but not for Type 1 – design must provide more prestress or use propping.*
- *Curvature due to shrinkage $\kappa = N e_s/(E_c I_c)$ (with E_{ce}) → extra deflection $\delta = \kappa L^2/8$.*

$$\kappa = 4.615\times10^7/(13846\times7.687\times10^9) = 4.3\times10^{-7}/mm\ ;\ \delta = \kappa L^2/8 = 4.3\times10^{-7}\times10^8/8 = 5.4\ mm \downarrow$$

Always add the differential shrinkage stresses to the un-propped / propped stage stresses before checking the permissible limits.

10. Horizontal Shear at the Interface

The slab and the beam must act together – the **horizontal shear** along the interface is resisted by bond, friction of the roughened surface and **shear connectors (stirrups projecting into the slab)**.

$$\tau_h = V A_s \bar{y}_s / (I_c\, b)$$

- V = shear force acting on the composite section (after composite action, i.e. stage-3 loads : superimposed dead + live) ; A_s = area of the slab ; $\bar{y}_s$ = distance of slab C.G. from composite C.G.
- b = width of the interface (top width of the precast beam).

Example 6

At the support of the example beam $V = 7 \times 5 = 35$ kN (imposed) + live load, say total $V = 60$ kN on the composite section.

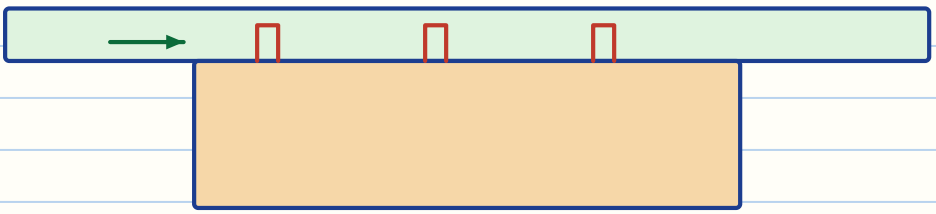
$$A_s \bar{y}_s = 100000 \times 166.7 = 1.667 \times 10^7 \text{ mm}^3$$

$$\tau_h = 60 \times 10^3 \times 1.667 \times 10^7 / (7.687 \times 10^9 \times 250) = 0.52 \text{ N/mm}^2$$

Compare with the permissible values (IS 456 / IS 1343 / IRC) for interface condition (roughened, with / without ties) – if higher, provide **vertical ties** : $A_v = \tau_h\, b\, s / (0.87\, f_y\, \mu)$.

Methods of providing composite action

- Roughen the top surface of the precast beam (≈ 5 mm amplitude), clean, wet.
- Projecting **stirrups** (shear keys) from the beam into the slab – spacing ≤ 300 mm / 4 × slab depth.
- Shear keys, indents, dowels, welded studs (steel–concrete composite).
- Adequate cover and anchorage of the ties in the slab and in the beam.



projecting stirrups act as shear connectors across the interface

Fig. Shear connectors (ties) at the beam–slab interface

11. Ultimate Strength of Composite Beams

At ultimate, the composite section is analysed like a PSC / RC T-section – the slab acts as the compression flange (neutral axis usually in the slab). Strain in steel includes the effective prestress strain.

- Compression $C = 0.36\, f_{ck}\, b_f\, x_u$ (f_{ck} of the slab) ; tension $T = A_{ps}\, f_{pu}$; $\Sigma H = 0 \rightarrow x_u$.
- $M_u = T\,(d - 0.42\, x_u)$; check $M_u \geq 1.5\, M_{service}$ (IS 1343 load factor 1.5 for dead + live).
- Stress f_{pu} from IS 1343 (Fig. 3 / design curves), depends on $A_{ps}/(b\, d)$, f_{ck} and effective prestress.

Example 7

For the above beam : $A_{ps} = 636$ mm² (≈ 6 strands 12.7 mm ; $P_i = 700$ kN), depth to tendon from top of slab $d = 100 + 250 + 130 = 480$ mm, f_{ck} (slab) = 40, assume $f_{pu} \approx 1400$ N/mm² (from IS 1343 curves), $b_f = 1000$ mm.

$$T = 636 \times 1400 = 890\ \text{kN} ; C = 0.36 \times 40 \times 1000 \times x_u = 14400\, x_u$$

$$x_u = 890 \times 10^3 / 14400 = 61.8\ \text{mm} < D_f = 100\ \text{mm} \checkmark \text{ (N.A. in slab)}$$

$$M_u = 890 \times (480 - 0.42 \times 61.8) = 890 \times 454 = \mathbf{404\ kN{\cdot}m}$$

$$M_{service} = 39.06 + 31.25 + 87.5 = 157.8\ \text{kN{\cdot}m} ; 1.5 \times 157.8 = 236.7\ \text{kN{\cdot}m} < 404 \checkmark \text{ safe}$$

Note : In un-propped construction the precast beam carries stage-2 moment alone; but at ultimate the composite section is assumed to act as a whole (as the beam and slab are connected) provided the interface shear is resisted.

Check also : (i) shear (V_{co}, V_{cr} with composite section), (ii) interface shear, (iii) deflection (including differential shrinkage curvature), (iv) cracking/fatigue if required.

12. Important Questions & Formula Sheet

Short answers

1. What is a composite beam ? State its advantages.
2. Differentiate between propped and un-propped construction.
3. What is differential shrinkage ? How is it handled in analysis ?
4. Define effective flange width of a T-beam (IS 456).
5. How is the horizontal shear between precast beam and slab resisted ?
6. What is the effect of creep on differential shrinkage stresses ?

Long answers

1. Explain the stages of loading and the stress distribution in a composite beam constructed (a) un-propped (b) propped. Compare them.
2. A composite beam consists of a PSC beam … Find the stresses at all stages (numerical – as Example 5).
3. Derive the stresses due to differential shrinkage in a composite beam. Numerical (Example 6).
4. Explain the design of shear connectors; compute the interface shear stress.
5. Explain the types of composite beams with sketches.

Formula sheet

Item	Formula
Stage 2 (precast alone)	$f = P_e/A_1 \mp P_e e/Z_1 \pm (M_{g1} + M_s)/Z_1$
Stage 3 (composite)	$f = M_q\, y / I_c$
Composite section	$\bar{y} = \Sigma Ay/\Sigma A$; $I_c = \Sigma I + \Sigma A\, d^2$
Effective flange width	$b_f = L_0/6 + b_w + 6D_f$
Modular ratio	$m = E_{slab}/E_{beam}$
Differential shrinkage force	$N = \varepsilon_{ds}\, E_{ce}\, A_s$, $E_{ce} = E_c/(1+\theta)$
Shrinkage stress	$f = N/A_c \pm N e_s\, y/I_c$ (release) $- N/A_s$ (slab)
Interface shear	$\tau_h = V A_s \bar{y}_s/(I_c\, b)$
Ultimate moment	$M_u = A_{ps} f_{pu} (d - 0.42\, x_u)$

Best wishes for your exams ! Practise the stage-wise stress tables – they carry most of the marks.